

- Q.28 Write difference between active and passive transducer. (CO2)
- Q.29 Define different types of pressure cells and their uses. (CO5)
- Q.30 Write short note on thermistors? (CO7)
- Q.31 Explain the working of Bimetallic thermometer. (CO7)
- Q.32 What are different methods of Measurement. (CO1)
- Q.33 Write short note on electromagnetic flow meter. (CO6)
- Q.34 Explain platinum resistance thermometer. (CO7)
- Q.35 What are the various types of hygrometers? Explain any one. (CO8)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain working and construction of capacitive transducer. (CO3)
- Q.37 Write a short note on:
a) thermocouple
b) RTD. (CO7)
- Q.38 What are the different ways according to which transducer can be classified? List some factors determining traducer. (CO2)

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Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following has a positive temperature coefficient? (CO7)
a) RTD b) Thermistor
b) Thermocouple c) Mercury thermometer
- Q.2 The transducer output is: (CO1)
a) Exponential b) Uni step
c) Non-linear d) Linear
- Q.3 Piezoelectric transducers work when we apply _____ to it. (CO2)
a) Heat b) Mechanical force
c) Illumination d) Vibrations
- Q.4 Strain gauge is a: (CO3)
a) Inductive transducer
b) Capacitive transducer
c) Resistive transducer
d) Mechanical transducer

- Q.5 What is the principle of operation of LVDT? (CO3)
- Mutual-inductance
 - Self-Inductance
 - Reluctance
 - Permanence
- Q.6 Load cell is used for the measurement of: (CO4)
- Area
 - Length
 - Mass
 - Force
- Q.7 Capacitive transducer is used for: (CO2)
- Static measurement
 - Dynamic measurement
 - Transient measurement
 - Both static & dynamic
- Q.8 Which of the following represent active transducer? (CO7)
- Strain gauge
 - Thermistor
 - LVDT
 - Thermocouple
- Q.9 Pressure is the: (CO5)
- Mass per unit area
 - Force per unit area
 - Force per unit volume
 - Mass per unit volume
- Q.10 LVDT is used to measure: (CO3)
- Force
 - Displacement
 - Temperature
 - Frequency

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Write full form of RTD. (CO7)
- Q.12 Define electrical transducer. (CO2)
- Q.13 Name any two methods of measure temperature. (CO7)
- Q.14 Define Gauge pressure. (CO5)
- Q.15 The unit of pressure is _____. (CO5)
- Q.16 Write full form of LVDT. (CO3)
- Q.17 What is the function of tachometer? (CO2)
- Q.18 Define hygrometer. (CO8)
- Q.19 What is load cell. (CO4)
- Q.20 Define error. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write advantages and disadvantages of LED. (CO1)
- Q.22 What is piezoelectric transducer? Write its application. (CO2)
- Q.23 What is the importance of measurement? (CO1)
- Q.24 Explain working principle of LVDT? (CO3)
- Q.25 What are different types of measurement of force? (CO4)
- Q.26 Write short note on “Proving Rings”. (CO4)
- Q.27 What is signal conditioning? Why is it needed? (CO1)